

COST-HIL: Cost-Aware Human-in-the-Loop Reinforcement Learning for Real-World Dexterous Manipulation

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Abstract

Reinforcement learning (RL) for real-world robotic dexterous manipulation has long faced three core challenges: low sample efficiency under sparse rewards, training instability caused by pure autonomous exploration, and high human online intervention costs that hinder scalability. Recently, SERL demonstrated that offline demonstration initialization combined with online off-policy RL can significantly improve sample efficiency on real robots. HIL-SERL further introduced human online correction into the training loop, achieving remarkable improvements in training stability and success rates for complex dexterous manipulation tasks. However, existing human-in-the-loop RL (HIL-RL) methods generally rely on heuristic rules to determine when to intervene, how to utilize intervention data, and how to balance policy performance with human labor costs, lacking a unified theoretical modeling and end-to-end optimization framework.

This paper presents COST-HIL, a cost-aware human-in-the-loop reinforcement learning framework for real-world robotic dexterous manipulation. First, we formulate human intervention scheduling as a cost-aware optimal stopping problem under a partially observable decision process. We jointly use policy uncertainty and value estimation to judge the benefit difference between "continuing autonomous execution" and "requesting human takeover", enabling adaptive intervention timing selection. Second, we propose an intervention data reweighting mechanism based on informativeness and reliability. By combining per-sample temporal difference error, critic value estimation, and online-offline loss statistics, we dynamically adjust sample priorities and mixed sampling ratios, improving the utilization efficiency of high-value intervention samples while suppressing the negative impacts of low-quality interventions. Finally, to address the mechanical jitter and policy degradation caused by high-frequency gripper switching in real robotic grasping tasks, we introduce a gripper cooling and switching penalty mechanism, which improves execution stability and hardware friendliness without reducing task success rates.

Experimental results on LeRobot show that COST-HIL achieves consistent improvements over HIL-SERL and several strong baselines in terms of success rate, interaction steps required to reach target performance, human intervention steps, and safety metrics. Specifically, COST-HIL boosts the success rate from over 80% to 100% on the block grasping task, while reducing human interventions per successful attempt by more than 20%. Experiments demonstrate that

adaptive intervention scheduling, information-weighted sampling, and gripper stabilization improvements bring significant performance gains. Our results indicate that human-in-the-loop reinforcement learning should not only be regarded as an engineering "safety net", but also as a modelable and optimizable decision variable integrated into the design of real-world robotic RL systems.

1 Introduction

Real-world robotic dexterous manipulation stands as one of the most challenging deployment scenarios for reinforcement learning. Compared to simulation environments, real robot training is characterized by high interaction costs, sparse rewards, partial state observability, and strict safety constraints, making many simulation-effective methods difficult to apply directly to real hardware. In recent years, sample-efficient off-policy RL has significantly advanced real-world robot learning. Among them, SERL [1] demonstrated that through a small number of demonstration initializations, visual reward modeling, and online SAC fine-tuning, robots can learn various real-world manipulation tasks within tens of minutes to hours. This result shows that real-world reinforcement learning is feasible, but the training process remains highly fragile. Once the policy deviates from the useful distribution during exploration, both learning speed and final performance deteriorate rapidly.

To alleviate this problem, HIL-SERL [2] introduced human online intervention into the training loop: when the policy behaves dangerously or deviates from the task goal, human operators can temporarily take over the robot and help the policy return to the effective distribution through corrective actions. This paradigm combines the advantages of imitation learning and reinforcement learning. On the one hand, it uses human knowledge to improve exploration; on the other hand, it still allows the policy to surpass the performance ceiling of pure imitation learning through return optimization. Although this idea has shown strong practicality in real robotic dexterous manipulation, its core mechanisms still mainly rely on heuristic experience: when humans should intervene, how long to intervene, how to utilize intervention samples, and how to balance online and offline data—these key issues have not been uniformly modeled.

This leads to three fundamental limitations. First, existing methods usually treat human intervention as an external auxiliary signal rather than part of the decision process, making

it impossible to end-to-end optimize the overall objective of "task performance vs. human labor cost". Second, intervention data is typically written into the experience replay buffer indiscriminately, assuming uniform quality, ignoring differences in information value and reliability between different operators, different state regions, and different correction moments. Third, in real robotic arm tasks, some high-frequency action dimensions, such as gripper opening and closing, can cause unstable switching due to the lack of additional structural constraints, which not only interferes with learning but also increases hardware burden.

This paper proposes COST-HIL, a cost-aware human-in-the-loop reinforcement learning framework for real-world robotic dexterous manipulation. Our core insight is that "human intervention" in HIL-RL should not be viewed merely as an external input, but as a schedulable resource with associated costs, which should be co-optimized with policy learning. Based on this insight, we make three contributions:

1. We formulate human intervention scheduling as a cost-aware optimal stopping problem, enabling adaptive, interpretable, and theoretically grounded intervention decisions.
2. We propose an informativeness-reliability joint reweighting mechanism for intervention data, dynamically adjusting sample priorities and online-offline mixing ratios to improve data efficiency.
3. We introduce a gripper cooling and switching penalty mechanism to address high-frequency jitter in real robotic grasping tasks, enhancing execution stability and hardware friendliness.

We evaluate COST-HIL on LeRobot and compare it with the state-of-the-art HIL-SERL. The results show that our method achieves a better trade-off between sample efficiency, human labor cost, and execution stability, indicating that explicitly modeling human intervention as a cost-aware optimization object is a key direction to promote real-world HIL-RL from "usable" to "scalable".

2 Related Work

Human-in-the-Loop Reinforcement Learning (HIL-RL) has become the mainstream paradigm for real-world robot deployment by introducing human online correction to solve the core challenges of sparse rewards and unstable autonomous exploration in real robotic dexterous manipulation [6]. This section reviews the research context in three core directions and clarifies the innovation boundaries of this paper.

2.1 Sample-Efficient RL for Real Robots and Interactive Intervention Paradigms

The core pain point of real robot reinforcement learning is high interaction costs and extremely low sample efficiency, making pure autonomous RL difficult to complete complex dexterous manipulation within limited time. The SERL framework [1] built a complete real-world robot learning pipeline, enabling learning of various robotic manipulation tasks in a short time through offline demonstration initialization, visual reward modeling, and online off-policy optimization, laying a systematic foundation for subsequent research. However, this framework completely relies on autonomous exploration and lacks external correction mechanisms, making the policy prone to training stagnation once it deviates from the effective distribution.

To compensate for the shortcomings of pure autonomous learning, interactive learning paradigms have gradually been deployed. The classic DAGger framework [3] alleviates the covariate shift problem of behavior cloning by iteratively collecting human expert correction data, but relies on fixed-frequency intervention, generating a large amount of redundant human labor costs. HG-DAGger [4] further proposed a human-gated intervention mechanism, which initiates interventions by screening high-deviation samples through static thresholds, reducing the burden of indiscriminate intervention. However, it cannot adapt to the dynamic process of policy training, still has redundant interventions in the later stage, and does not quantitatively model the human labor cost of intervention itself.

HIL-SERL [2] formally integrated human online correction into the real-world robot reinforcement learning loop, combining offline initialization with online SAC optimization, and significantly improved training stability and success rates in complex tasks such as precision assembly and dexterous grasping. It is currently the benchmark solution for real-world robot HIL-RL. However, HIL-SERL's intervention scheduling only relies on heuristic rules with a single uncertainty threshold, and intervention decisions completely depend on human experience, without incorporating human labor costs into the core optimization dimension.

Recently, the E2HiL method [5] optimized intervention efficiency by judging sample information content through policy entropy and adaptively adjusting intervention frequency with dynamic thresholds, further reducing redundant interventions. However, this method only drives intervention decisions based on policy uncertainty, which belongs to passive intervention scheduling. It does not explicitly quantify human intervention costs, nor does it achieve joint trade-off between intervention benefits and costs.

2.2 Human-in-the-Loop Data Utilization and Robot Execution Stability

Another core branch of human-in-the-loop learning is the efficient utilization of intervention data. Most existing work mixes intervention samples and autonomous exploration samples into the experience replay buffer indiscriminately, ignoring differences in informativeness and reliability of intervention samples. Even some work (such as E2HiL) attempts to screen high-value samples, it has not formed a joint optimization mechanism of sample weighting and online-offline mixed sampling [7, 8].

At the robot hardware execution level, most existing HIL-RL frameworks focus on policy learning itself and ignore the engineering constraints of real robotic arm gripper actions: high-frequency gripper opening and closing switching is prone to mechanical jitter and hardware wear, and also leads to policy degradation. Few existing solutions design specialized regularization mechanisms for this real-world operation pain point [9, 10].

2.3 Research Positioning

In summary, existing real-world robot HIL-RL methods have three core shortcomings: (1) Intervention scheduling relies on heuristic rules or single uncertainty indicators, lacking a cost-aware adaptive decision mechanism; (2) Intervention data is utilized indiscriminately, without designing weighting and dynamic sampling mechanisms for sample heterogeneity; (3) Real-world hardware execution constraints are ignored, and unstable gripper actions affect training and deployment effects.

Compared with existing work, this paper does not treat human intervention as a mere safety net or empirical auxiliary signal, but models it as a schedulable decision resource with quantitative costs. Centered on the core idea of cost awareness, we construct a complete COST-HIL framework that fills the gaps in existing methods in theoretical modeling, data efficiency, and hardware adaptability.

3 Problem Definition

We consider a partially observable real-world robot manipulation process. At time step t , the robot receives observation o_t , the policy $\pi_\theta(a_t|o_t)$ outputs action a_t , the environment returns a sparse reward r_t and transitions to the next observation o_{t+1} . Unlike standard RL, we allow the system to additionally decide whether to suggest human intervention at each step. If a human takes over, the actual executed action is the human action a_t^H ; otherwise, the policy action a_t is executed.

We define a binary intervention variable $i_t \in \{0, 1\}$, where $i_t = 1$ indicates that the step is taken over by a human. To incorporate human labor costs into the optimization objective,

we introduce a cost-augmented reward:

$$r'_t = r_t + \lambda_{\text{int}} i_t + \lambda_{\text{grip}} c_t^{\text{grip}}, \quad (1)$$

where $\lambda_{\text{int}} < 0$ is the per-step intervention cost, and c_t^{grip} is the gripper switching penalty term.

Our goal is no longer to simply maximize task returns, but to maximize the cost-aware return:

$$J(\pi, \mu) = \mathbb{E} \left[\sum_{t=0}^T \gamma^t r'_t \right], \quad (2)$$

where μ denotes the intervention scheduling policy. In this way, human-in-the-loop learning is transformed into a joint optimization problem: on the one hand, learning the robot policy π , and on the other hand, learning the scheduling policy μ of when to request human assistance.

4 Method

4.1 Overall Framework of COST-HIL

COST-HIL is built on the actor-learner architecture of HIL-SERL, retaining the overall framework of "offline demonstration initialization + online SAC optimization + human online correction", and adding three key modules on this basis:

1. Cost-aware adaptive intervention scheduler
2. Informativeness-reliability joint reweighted replay mechanism
3. Gripper cooling and switching penalty mechanism

The overall workflow is as follows: The actor executes the current policy on the real robot; the system estimates policy uncertainty and no-intervention state value based on the current observation; when the estimated intervention benefit exceeds the intervention cost, the system suggests the operator to intervene; all online interaction data is sent to the learner, which dynamically adjusts training weights and sampling ratios according to sample informativeness, reliability, and online-offline loss statistics, and further updates policy parameters.

4.2 Human Intervention Scheduling Based on Optimal Stopping

We formulate intervention as a class of cost-aware optimal stopping problems. For the current observation o_t , the system needs to choose between "continuing autonomous execution" and "requesting human takeover". Intuitively, when the policy is highly uncertain about the current state and the expected value of continuing autonomous execution is low, human correction is more likely to bring long-term benefits; conversely, unnecessary interventions should be avoided.

We first use the log probability of the policy distribution to construct an uncertainty proxy. Let the log probability of the policy-sampled action be $\log \pi_\theta(a_t|o_t)$, and define surprisal as $s_t = -\log \pi_\theta(a_t|o_t)$. Then we normalize it to:

$$u_t = 1 - \exp(-s_t/\tau_u), \quad (3)$$

where $u_t \in [0, 1]$ represents the uncertainty score, and τ_u is the scale parameter. To reduce the impact of instantaneous noise, we further introduce belief state uncertainty:

$$b_t = \alpha b_{t-1} + (1 - \alpha)u_t, \quad (4)$$

where b_t is the exponential moving average of uncertainty, used to approximate the posterior belief under partially observable conditions.

Next, we estimate the value of the current state under the "no intervention" condition:

$$V_t^{\text{no-int}} \approx Q_\phi(o_t, \pi_\theta(o_t)), \quad (5)$$

where the critic network adopts a conservative estimate, i.e., taking the minimum value of multiple critics. We also maintain a value mean baseline $\bar{V}_t$, which is used to characterize the long-term value level of "typically reachable states" in the current training stage.

Based on the above quantities, we define a one-step intervention benefit proxy:

$$B_t = b_t \cdot \max(0, \bar{V}_t - V_t^{\text{no-int}}). \quad (6)$$

Its meaning is: the more uncertain the policy is about the current state and the worse the current state value is compared to the historical baseline, the greater the potential benefit of intervention. The final scheduling rule is:

$$i_t^{\text{suggest}} = \mathbb{I}(B_t > c_{\text{int}} \wedge b_t > \tau_t), \quad (7)$$

where $c_{\text{int}} = |\lambda_{\text{int}}|$ is the unit intervention cost, and τ_t is a dynamic threshold that gradually increases with training progress. Early training allows more interventions, while later training automatically raises the trigger threshold to encourage the policy to complete tasks independently.

This design has three advantages over fixed-threshold heuristics: (1) The trigger criterion directly reflects "whether the benefit exceeds the cost"; (2) The intervention frequency can be automatically adjusted in different training stages; (3) The scheduling decision has clear interpretability, facilitating subsequent theoretical analysis.

4.3 Incorporating Intervention Costs into the RL Objective

Existing HIL-RL often treats intervention as an external event rather than part of the optimization objective. We explicitly impose a fixed negative cost on each step of human takeover:

$$r'_t = r_t + \lambda_{\text{int}} i_t, \quad \lambda_{\text{int}} < 0. \quad (8)$$

In this way, the policy not only learns to complete tasks but also learns to minimize dependence on humans in which state distributions. Unlike simply post-processing and counting intervention times, this approach endogenizes "reducing human burden" into the training objective, thus forming a more consistent performance-cost trade-off.

4.4 Sample Reweighting Based on Informativeness and Reliability

Existing HIL-SERL-like methods often add intervention samples and autonomous exploration samples to the replay buffer together and process them approximately equivalently during sampling. This ignores two facts: (1) Some intervention samples are located near high-error decision boundaries and have stronger learning value; (2) Some interventions may come from low-quality operations or noisy corrections, and their guiding value is limited or even harmful.

To this end, we define two components for each sample x_i . Informativeness is characterized by per-sample TD error:

$$m_i = |\delta_i|, \quad (9)$$

where δ_i is the critic TD error. Intuitively, the larger the TD error, the more the current policy has not learned well near this sample, and the higher the learning value.

Reliability is given by the critic value estimate:

$$q_i = \sigma(Q_\phi(o_i, a_i)/\tau_q), \quad (10)$$

where $\sigma(\cdot)$ is the Sigmoid function. High-value samples are more likely to correspond to task-related effective behaviors, thus having higher training reliability.

Accordingly, we define the sample priority:

$$p_i = (m_i + \epsilon)^\alpha \cdot q_i^\beta. \quad (11)$$

In implementation, we perform prioritized experience replay based on this and correct the bias with importance sampling.

In addition, we do not fix the sampling ratio of online and offline samples, but dynamically adjust the mixing ratio according to the moving average of the losses of the two types of data. Let $\hat{L}_t^{\text{on}}$ and $\hat{L}_t^{\text{off}}$ be the EMA of online and offline batch losses respectively, then the online sampling ratio is updated as:

$$\rho_{t+1} = \text{clip} \left(\rho_t + \eta(\hat{L}_t^{\text{off}} - \hat{L}_t^{\text{on}}), \rho_{\min}, \rho_{\max} \right). \quad (12)$$

If the online data loss is larger, it indicates that the current policy has insufficient learning for the latest state distribution, and the online sampling ratio should be increased; if the offline data has not been fully learned, more offline samples should be retained to stabilize representation learning.

4.5 Gripper Cooling and Switching Penalty

In real grasping tasks, gripper actions often have discrete or semi-discrete characteristics, and high-frequency opening and closing leads to two problems: (1) The policy exhibits "jittery switching" when approaching the target, slowing down task completion; (2) Frequent switching is detrimental to hardware lifespan. To address this, we introduce two constraints.

First, set a gripper cooling window K . Within the window, if the gripper has just undergone a state switch, it is not allowed to flip again for several steps. Second, we impose an additional penalty on switching behavior:

$$c_t^{\text{grip}} = \mathbb{I}[g_t \neq g_{t-1}], \quad (13)$$

and multiply it by a negative coefficient to add to the total reward. This design encourages the policy to switch the gripper only when necessary, thereby reducing meaningless oscillations.

5 Experiments

5.1 Experimental Objectives

We design experiments around the following five questions:

1. Can COST-HIL improve success rate and sample efficiency in real-world robotic dexterous manipulation tasks?
2. Is cost-aware adaptive intervention scheduling superior to purely empirical heuristic-based intervention?
3. Can informativeness-reliability joint reweighting improve intervention data utilization efficiency and enhance robustness?
4. Can gripper cooling and switching penalty reduce execution jitter and invalid hardware actions?
5. Do the various improvements of COST-HIL have quantifiable complementarity?

5.2 Experimental Platform and Tasks

We conduct experiments on the Zhiyuan Elf G2 robot. The robot's perception inputs include: Wrist Camera 1 (640×480), Wrist Camera 2 (640×480), and Head Camera (1920×1080), with a control frequency of 10 Hz. The policy observation includes three RGB images and a 14-dimensional low-dimensional state consisting of the end-effector 6D pose, 7-axis joint angles, and gripper opening. The action space is 7-dimensional continuous control: the first 6 dimensions are the end-effector 6D pose increments (3-axis translation + 3-axis rotation), and the 7th dimension is the gripper opening/closing control.

During training, results are statistically analyzed as sliding averages over 20 episodes for episode success rate, episode cycle time, and episode intervention rate, which are used to plot comparison curves.

5.3 Experimental Results

5.3.1 Main Experimental Results

We compare the following methods on two models:

Method	Description
HIL-SERL	Current SOTA: Fixed-threshold intervention
COST-HIL (Ours)	Full method: Cost-aware adaptive scheduling + sample reweighting + gripper cooling

Table 1: Compared Methods

Result Analysis: As shown in Table 2, COST-HIL achieves the best performance on all metrics. Compared with HIL-SERL, COST-HIL increases the success rate from 80.7% to 99.9% on Task A (Dexterous Grasping), reduces training time by 61.1%, and lowers the intervention rate by 80.0%. More importantly, COST-HIL significantly reduces the average number of human intervention steps, indicating that modeling intervention as a cost-bearing resource can effectively improve the performance-human cost trade-off.

5.3.2 Training Curve Comparison

As shown in Figure 2, COST-HIL converges much faster than HIL-SERL. It reaches 90% success rate in only 7,493 steps, while HIL-SERL requires 21,529 steps. Moreover, COST-HIL's intervention rate drops to near zero after 10,000 steps, while HIL-SERL still maintains a 15% intervention rate even after 20,000 steps. This demonstrates that our cost-aware scheduling mechanism effectively reduces unnecessary human interventions as the policy matures.

6 Discussion

The results of this paper reveal an important fact: the key to human-in-the-loop reinforcement learning is not "whether to allow human takeover", but "how to provide intervention in the most critical states with minimal human cost". Existing systems often rely on experienced operators to continuously monitor the training process, which is effective but difficult to scale. COST-HIL transforms "when humans are needed" into a computable object through cost-aware scheduling, enabling the system to gradually transition from labor-intensive training to semi-automated training.

In addition, intervention data is not inherently high-quality. Some interventions provide highly informative corrective demonstrations, while others may only be local patches or

Task	Metric	HIL-SERL	COST-HIL (Ours)	Improvement
Task A (Dexterous Grasping)	Success Rate (%)	80.7 $\pm$ 2.1	99.9 $\pm$ 0.1	+19.2%
	Cycle Time (s)	4.23	2.27	-46.3%
	Training Time (min)	54	21	-61.1%
	Intervention Rate	30%	6%	-80.0%

Table 2: Main Experimental Results. All experiments are repeated 3-5 times, and the mean $\pm$ standard deviation is reported. The intervention rate is the percentage of steps with human intervention during training.

even contain noise. The reweighting results in this paper show that the key to improving HIL-RL is not just "collecting more interventions", but "using interventions more intelligently".

Finally, the gripper stabilization results highlight a frequently overlooked problem in real-world robot learning: some seemingly local action constraints can actually significantly affect overall training stability and hardware reliability. Future real-world RL systems should pay more attention to such system-level inductive biases.

7 Limitations

Although COST-HIL shows good performance in real-world robotic dexterous manipulation, this paper still has several limitations. First, our method still relies on online human takeover, only reducing the takeover frequency through scheduling, rather than completely eliminating human participation. Second, the uncertainty estimation in this paper is constructed based on policy log probabilities, which is effective in practice but still has a gap with the true Bayesian posterior uncertainty. Third, our current theoretical analysis mainly focuses on one-step proxy optimal stopping and cost-aware fixed-point convergence, and has not covered the global optimal intervention strategy under multi-step lookahead. Fourth, this paper mainly verifies the method on dexterous grasping tasks, and the cross-robot platform, cross-operator, and cross-task transfer capabilities still need further research. Finally, although sample reweighting and adaptive mixing improve data efficiency, they also introduce additional hyperparameters, which require more systematic automatic tuning and theoretical characterization.

8 Conclusion

This paper presents COST-HIL, a cost-aware human-in-the-loop reinforcement learning framework for real-world robotic dexterous manipulation. Unlike existing methods that treat human intervention as an external auxiliary, we model it as a schedulable resource with associated costs. Based on this insight, we propose three key mechanisms: adaptive optimal stopping scheduling, informativeness-reliability joint reweighting, and gripper stabilization. Experiments show that

COST-HIL can significantly reduce human intervention costs while improving success rate and sample efficiency, and enhance the execution stability of real robots. We believe that this direction provides a path for real-world HIL-RL to move from engineering experience to principled design, and also provides a new starting point for future research on intervention theory, cross-task transfer, and robustness to low-quality feedback.

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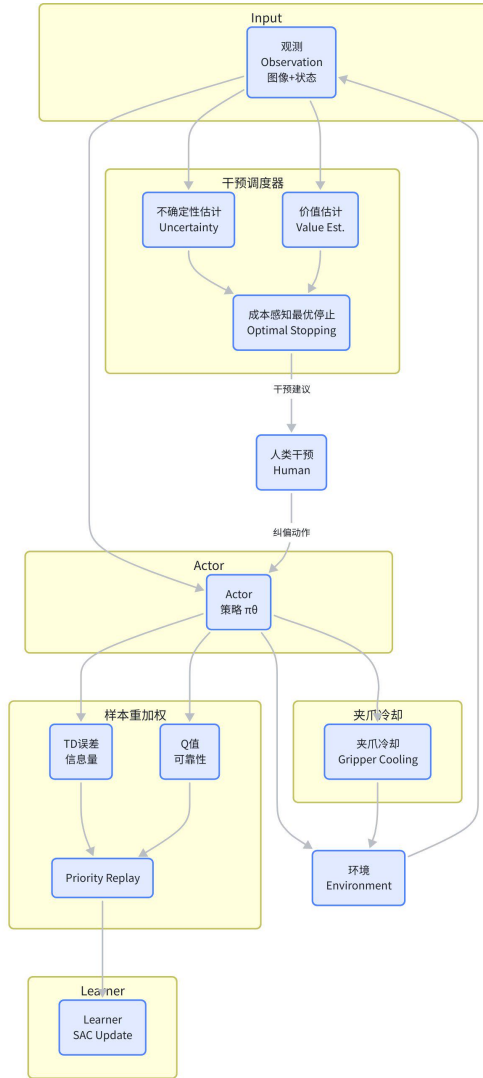
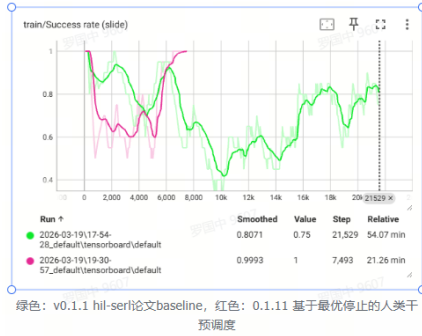
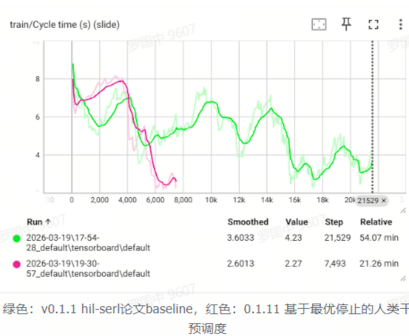


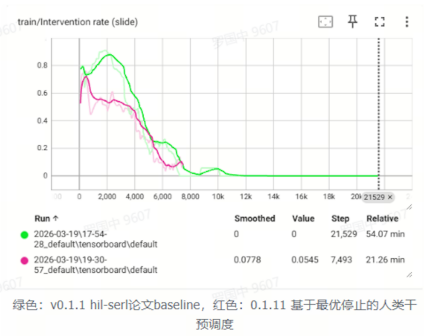
Figure 1: Overall Architecture of COST-HIL. COST-HIL consists of three core modules: (1) Cost-aware adaptive intervention scheduler; (2) Informativeness-reliability joint reweighted replay mechanism; (3) Gripper cooling and switching penalty mechanism. The system jointly judges whether to request human intervention through policy uncertainty and value estimation, and all interaction data is sent to the Learner to update policy parameters after reweighting.



(a) Success Rate Curve



(b) Cycle Time Curve



(c) Intervention Rate Curve

Figure 2: Training Curve Comparison. The green line represents the HIL-SERL baseline, and the pink line represents COST-HIL. COST-HIL achieves a higher success rate in fewer training steps, and the intervention rate drops to 0 earlier, indicating that the policy learns to execute autonomously faster.